# Kalman filter - for nonlinear systems
# Execise

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Given the following scalar system

$$x_k = a\sin(x_{k-1} + \varphi_f) + bu_{k-1} + w_{k-1} \tag{1a}$$
$$z_k = \sin(cx_k + \varphi_h) + v_k \tag{1b}$$
$$w \in \mathrm{NID}(0, Q)\ ,\ v \in \mathrm{NID}(0, R)\ ,\ x_0 \in \mathrm{N}(\hat{x}_0, P_0) \tag{1c}$$

assume $a, b, \varphi_f, c, \varphi_h, Q, R, x_0, P_0$ to be know.

A simple linear approximation of the non linear system (1) can be obtained by ignoring the nonlinear parts by approximating $\sin(x) \sim x$ and setting $\varphi_f = \varphi_h = 0$ to give the linear system (2).

$$x_k = ax_{k-1} + bu_{k-1} + w_{k-1} \tag{2a}$$
$$z_k = cx_k + v_k \tag{2b}$$

1. For the NL system (1), write the algorithm for the extended Kalman Filter.
2. The system can be simulated with the program NLSim.m. Here parameters are also given.
3. Try to program the EKF and make it work. If you don't succeed use the one in EKF.m. For comparision a linear KF for the system (2) is found in KF.m. To run these programs without errors you also need XCorrtk.m.
4. Compare and explain the performance of the EKF and linear KF for:
   (a) Initial parameters.[1]

   $$a = 0.95\ ,\ k = 1\ ,\ b = k(1-a)\ ,\ c = 1\ ,\ \varphi_f = 0\ ,\ \varphi_h = 0\ ,\ f_u = 0.02$$

   (b) $\varphi_h = \frac{\pi}{16}$.
   (c) $\varphi_f = \frac{\pi}{16}$.
   (d) $c = 10$.

[1] $f_u$ is the frequency for the square wave used as input in the simulation in NLSim.m.